UI/UX Project Design Report

Team: Design for Social Devastation

We are building an accessibility-focused classroom captioning and learning assistant interface designed to support real-time comprehension, retention, and review for students. The web dashboard provides live speech-to-text transcription during lectures, instantly accessible summaries, searchable full transcripts, and organized archives of past sessions.

Problem Statement & Purpose

- **Core Problem:** Students with hearing impairments, auditory processing differences, or non-native language backgrounds often struggle to keep up with fast-paced spoken lectures. Traditional note-taking further distracts from real-time engagement.

- **Solution:** ClassAccess delivers low-latency captioning paired with automated key-point extraction and archival tools, standardizing inclusive learning environments.

Core Module Analysis

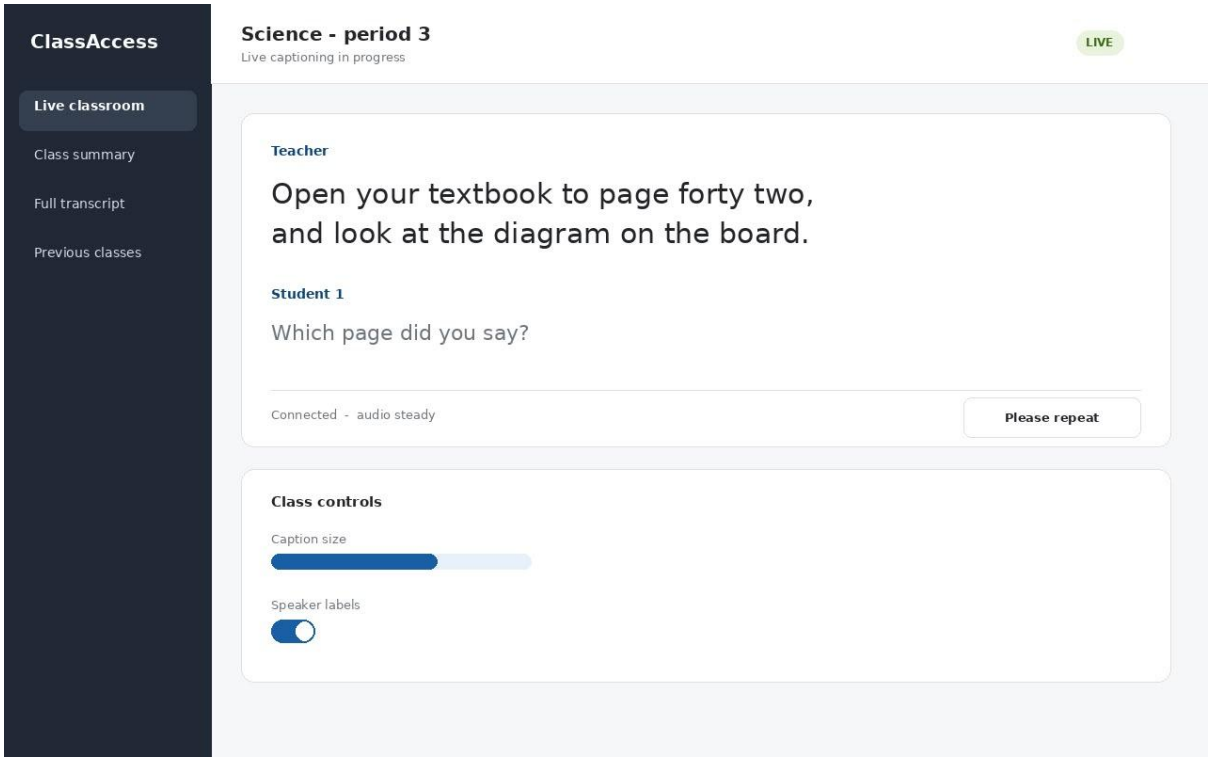
1. Live Classroom Interface

- **Real-Time Stream:** Displays live transcription segmented by speaker roles (e.g., *Teacher*, *Student 1*).

- **System Health Indicators:** Includes connection status (Connected - audio steady) and live status badges to build user confidence in real-time performance.

- **Interactive Assistive Button:** A prominent Please repeat button lets students flag missed audio discreetly without interrupting the class.

- **On-the-Fly Accessibility Controls:** Allows students to adjust **Caption Size** and toggle **Speaker Labels** live according to visual preferences.

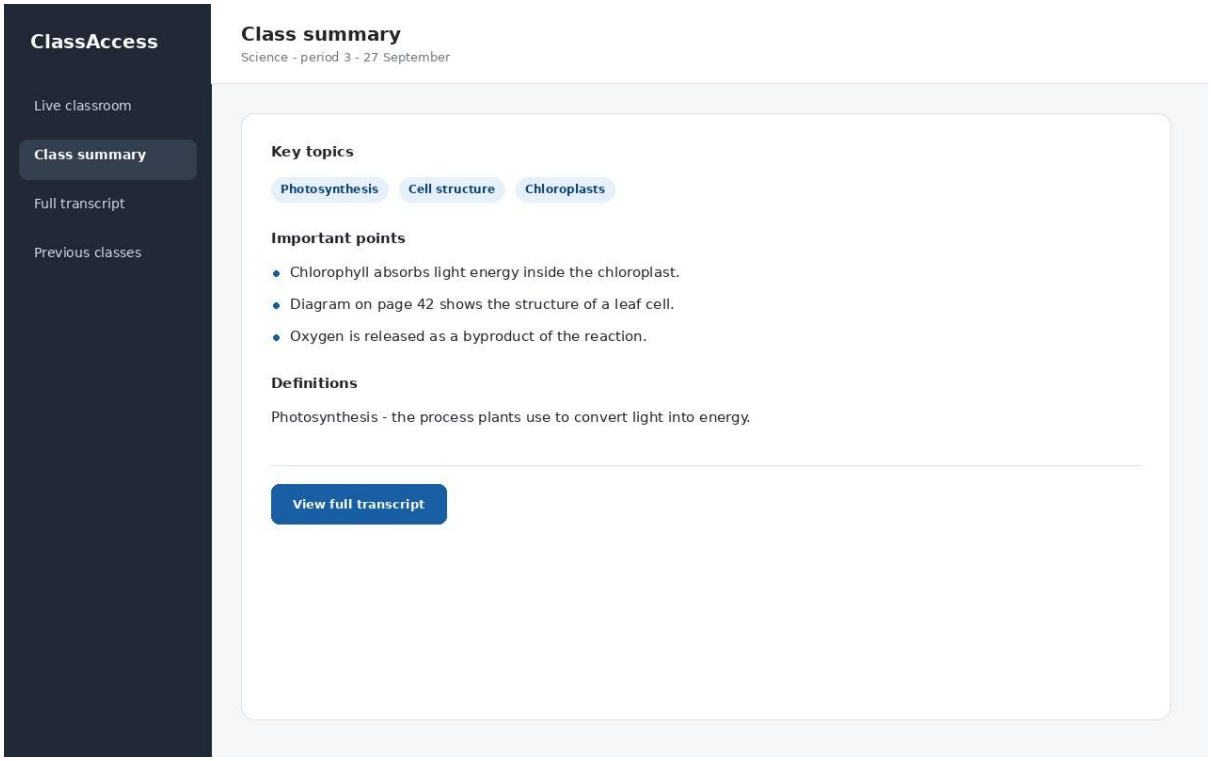


2. Class Summary Screen

- **Key Topic Tags:** High-level thematic pills (e.g., *Photosynthesis*, *Cell structure*, *Chloroplasts*) for rapid mental framing.

- **Structured Insights:** Automated extraction of core takeaway points and bolded key terminology definitions.

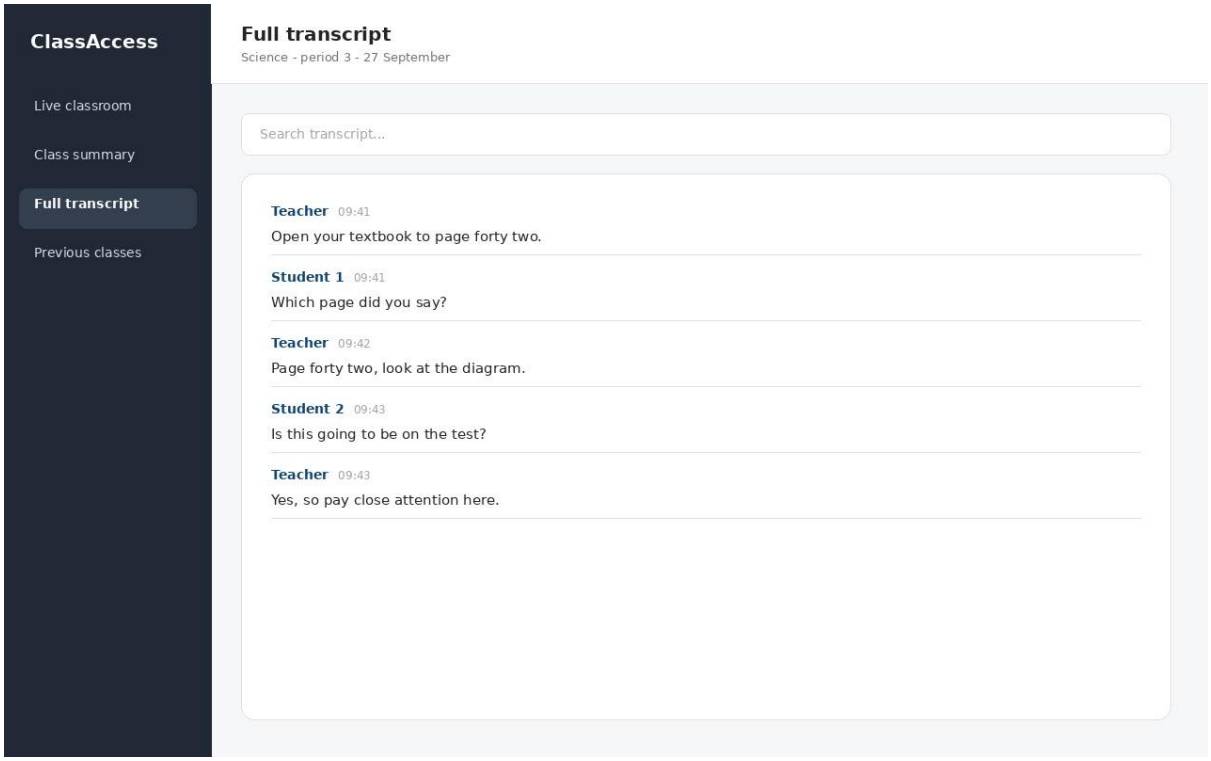
- **Direct Deep-Dive CTA:** A unified View full transcript button transitions students smoothly from high-level summaries to precise verbatim records.



3. Full Transcript Module

- **Search & Filter Bar:** Top-level input (Search transcript...) allowing fast navigation to specific terms, formulas, or teacher quotes.

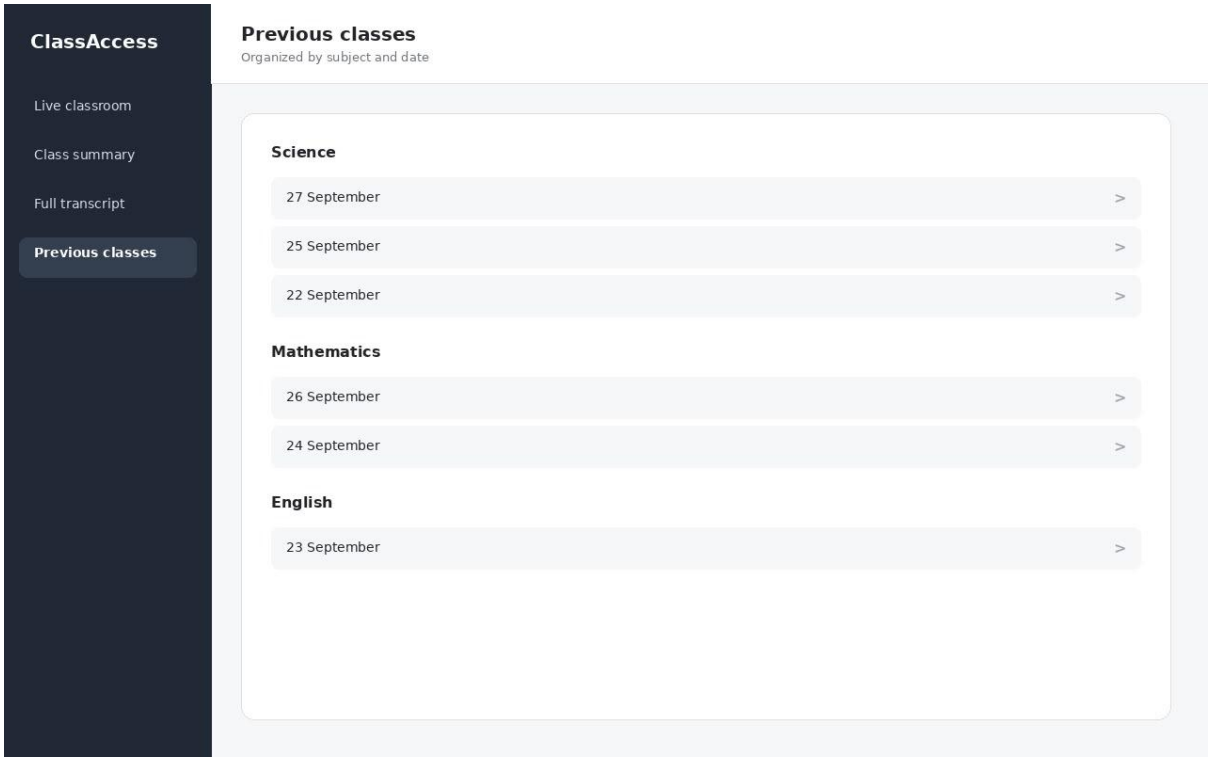
- **Timestamped Dialogue:** Complete sequential breakdown featuring speaker attribution and exact timestamps (e.g., 09:41).



4. Previous Classes Archive

- **Categorized Navigation:** Grouped cleanly by academic discipline (*Science*, *Mathematics*, *English*) and chronological date.

- **Consistent Layout:** Chevron-guided list cards enable quick navigation back into past summaries and transcripts.



Design System Specifications

Element	Specification / Standard	Design Rationale
Color Palette	Dark Slate (#1E2430), Accent Blue (#1D63B8), Soft Gray (#F5F6F8)	High contrast, low eye fatigue, WCAG 2.1 AAA alignment.
Typography	Clean Sans-Serif with hierarchical weight scales	Scalable rendering for large display captioning and small UI metadata.
Layout Structure	Fixed Left Navigation Bar + Fluid Main Content Canvas	High spatial predictability reduces cognitive load during live lectures.

Link:

https://www.figma.com/proto/BpyM3ODLBGOcJ0hG8YTs6H/DSI?node-id=13-84&viewport=-1535%2C861%2C0.61&t=GWWb0AGKgajrMfTa-1&scaling=min-zoom&content-scaling=fixed&starting-point-node-id=9%3A352&page-id=0%3A1